

01 BACKGROUND

Nonmedical prescription opioid use (NMPOU) is a serious adolescent health concern. Youth who misuse prescription opioids are markedly more likely to engage in **suicidal behavior, violence, and other substance use**, and many perceive these medications as safer than illicit drugs.

Prior research documents many risk factors for NMPOU — **but the same adversities also predict alcohol, marijuana, and other drugs**, so which are specific to opioids remains unknown.

96.6%

of all opioid misuse is prescription pain relievers

1 in 4

high-school seniors have used a prescription opioid

11.6%

lifetime NMPOU in this national sample

RESEARCH QUESTION

Which adversities create **opioid-specific hazard**, and which reflect **general substance-use liability**?

02 THREE FRAMEWORKS, THREE PREDICTIONS

F1 Generalized Risk-Taking NMPOU reflects a shared tendency toward reward-seeking, with no pathway to any one drug.

PREDICTS polysubstance use is the strongest predictor; **no adversity is opioid-specific**.

F2 Distress Relief Opioids specifically relieve emotional suffering (the self-medication hypothesis).

PREDICTS distress — **bullying, suicidality** — binds to opioids **more than to other drugs**.

F3 Contextual Adversity NMPOU arises from cumulative environmental adversity (sexual victimization, household, community).

PREDICTS these **raise substance use broadly, not opioids specifically**.

03 METHODS

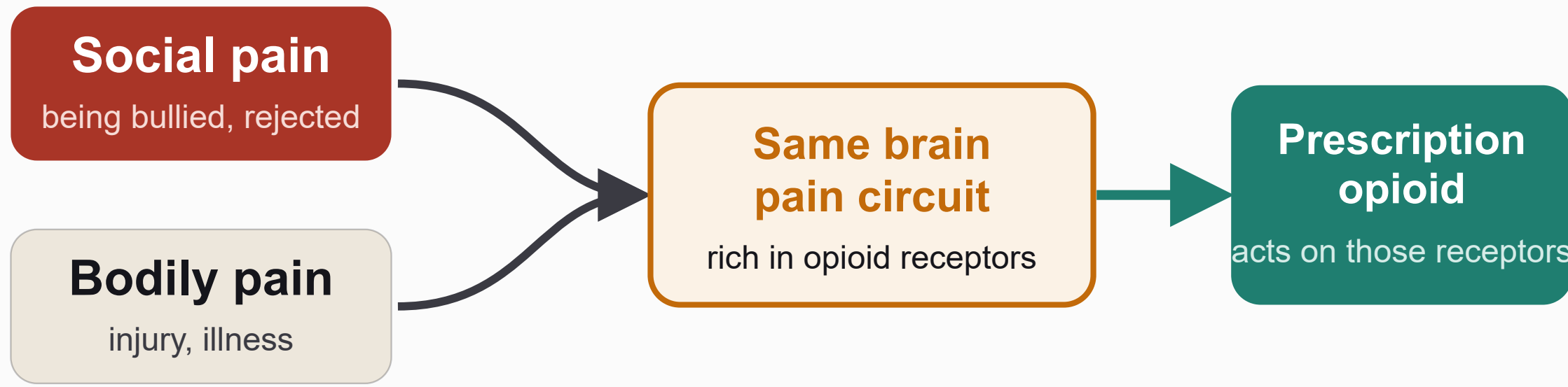
N 2023 Youth Risk Behavior Survey — nationally representative U.S. high-schoolers (grades 9-12), N = 20,103. Seven composite predictors span the three frameworks; survey-weighted models.

Stacked-outcome test. We model the predictors against **five drugs at once (opioids, alcohol, marijuana, cigarettes, cocaine)** in one analysis, then test whether a predictor's link is **stronger for opioids than for the others** — a direct test of opioid specificity.

% KHB decomposition measures how much of each adversity's opioid link is carried by general polysubstance use (lower = more direct).

04 WHY OPIOIDS, SPECIFICALLY

Social Pain Overlap Theory (SPOT)



The evidence: drugs that block these receptors make people feel less socially connected — so opioids relieve social pain in particular, which is why bullying may push youth toward opioids specifically.

Bullying tracks opioid use -- and no other drug.

A direct, opioid-specific "social pain" pathway that broad substance-use prevention misses.

+3.8 pp

higher opioid-use probability with bullying (AOR 1.48)

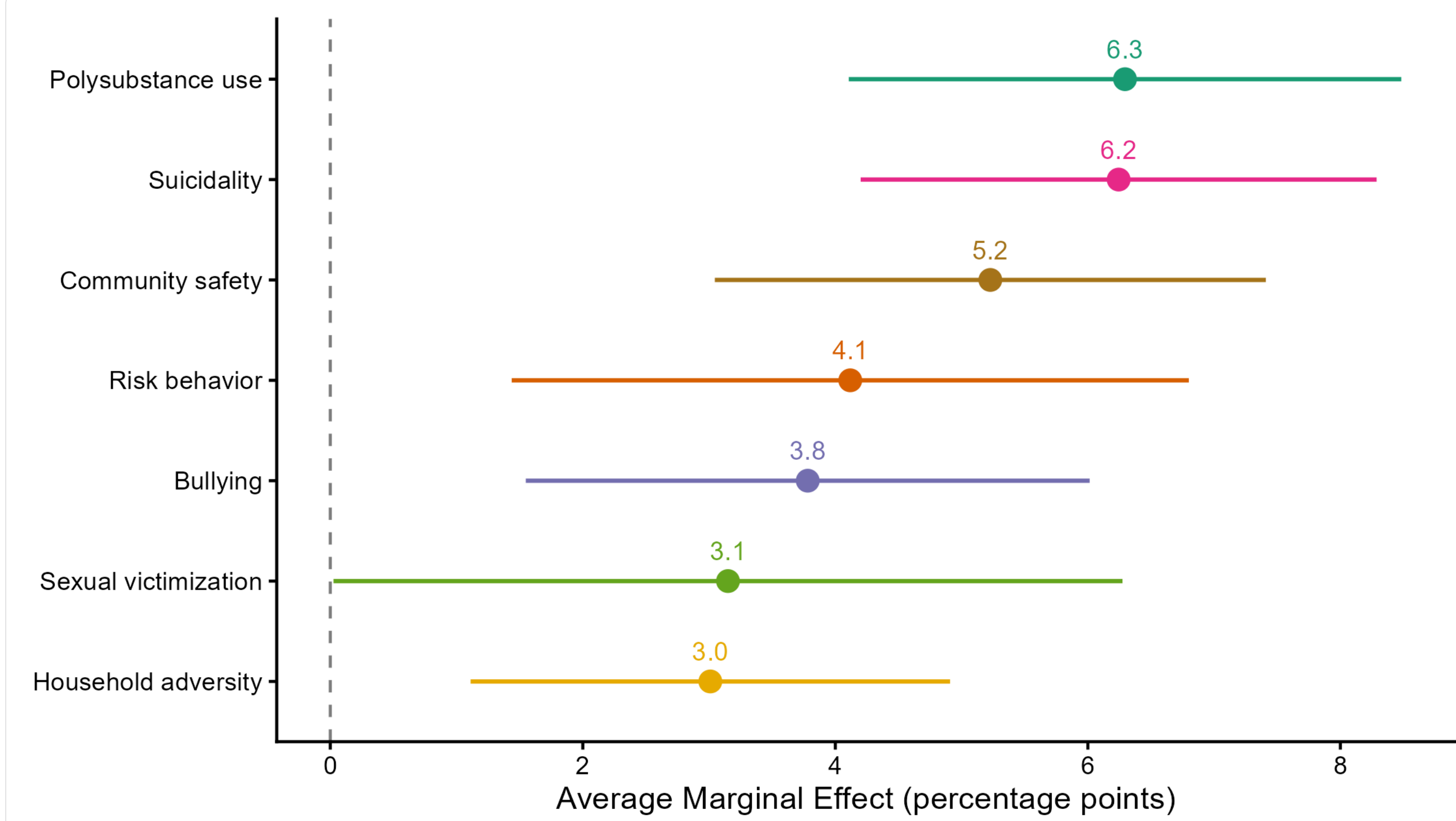
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other drugs bullying predicts (alcohol, MJ, cigarettes, cocaine)

10.3%

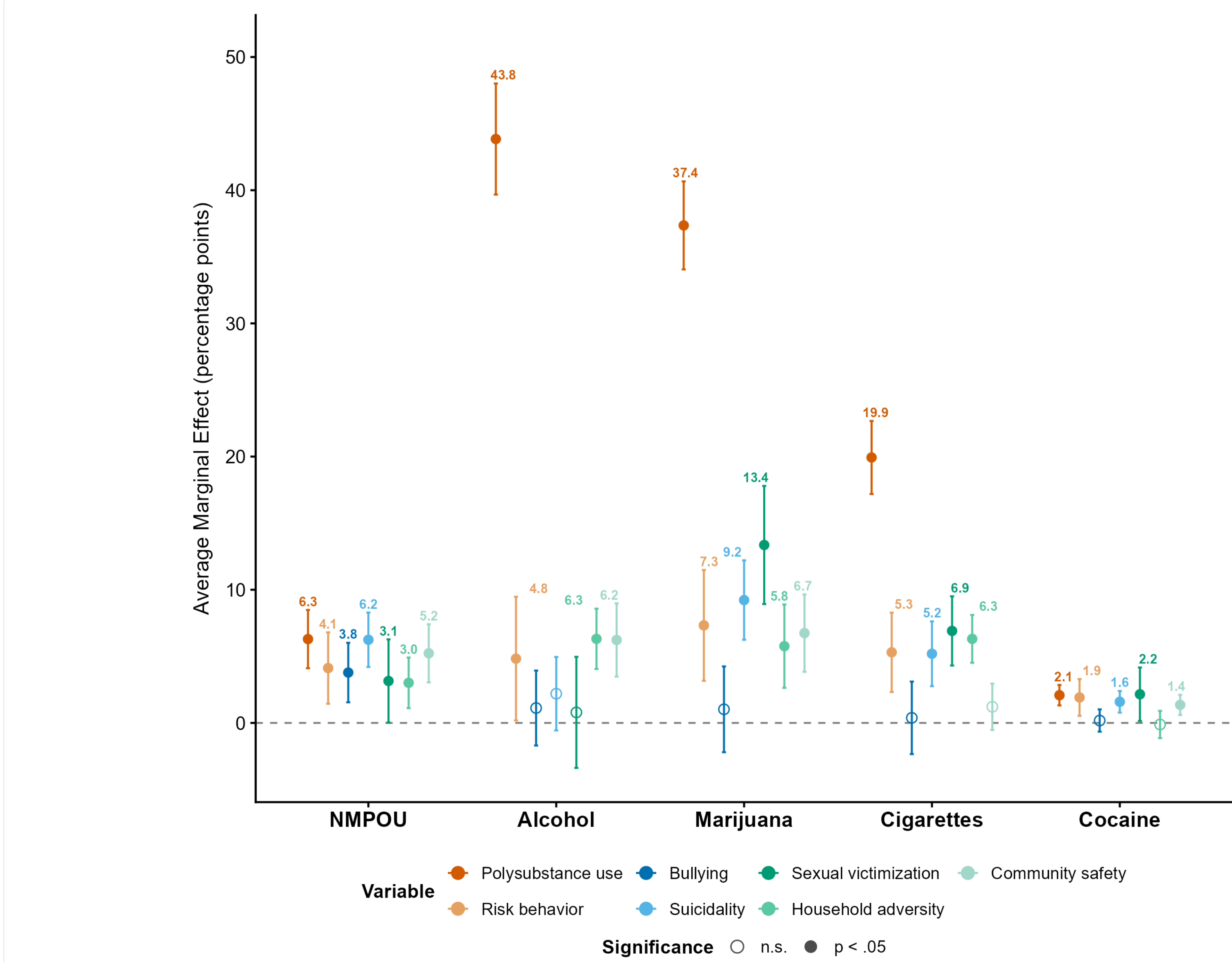
of the bullying-opioid link runs via other substances

STEP 1 EVERY ADVERSITY RAISES OPIOID RISK



All seven predictors are independently significant for opioids (n = 9,401); **polysubstance use is largest (6.3 pp)**. But magnitude alone cannot separate opioid-specific risk from general liability — so we test each predictor across five drugs.

STEP 2 BUT ONLY BULLYING IS OPIOID-SPECIFIC



Bullying clears significance for **opioids alone** and is flat for alcohol, marijuana, cigarettes, and cocaine. **Polysubstance use** is large everywhere = general liability; suicidality is opioid + marijuana, not alcohol = partial. Filled = p<.05; open = n.s. Heterogeneity across drugs: F = 4.61, p = 0.0075.

F1 GENERAL

Polysubstance largest (6.3 pp) — but large for every drug.

F2 OPIOID-SPECIFIC

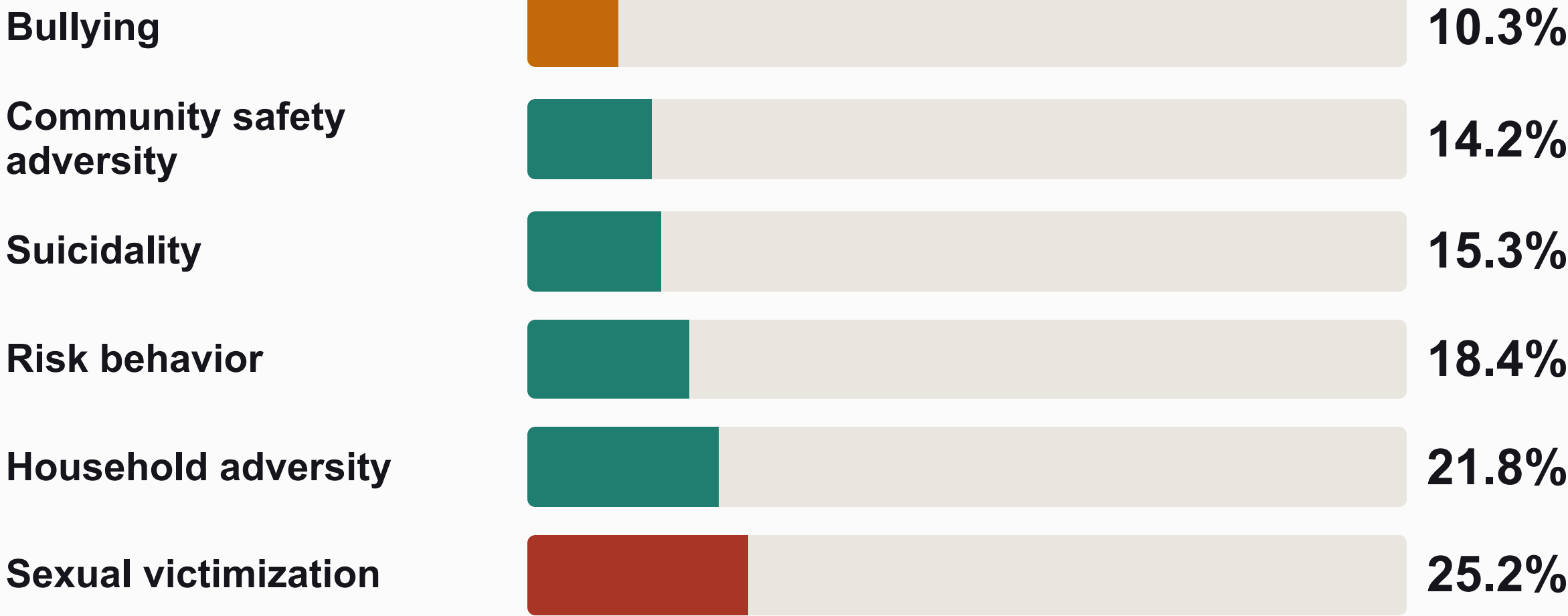
Bullying +3.8 pp, opioids only. Suicidality partial.

F3 GENERAL

Sexual victimization a hybrid; household & community broad.

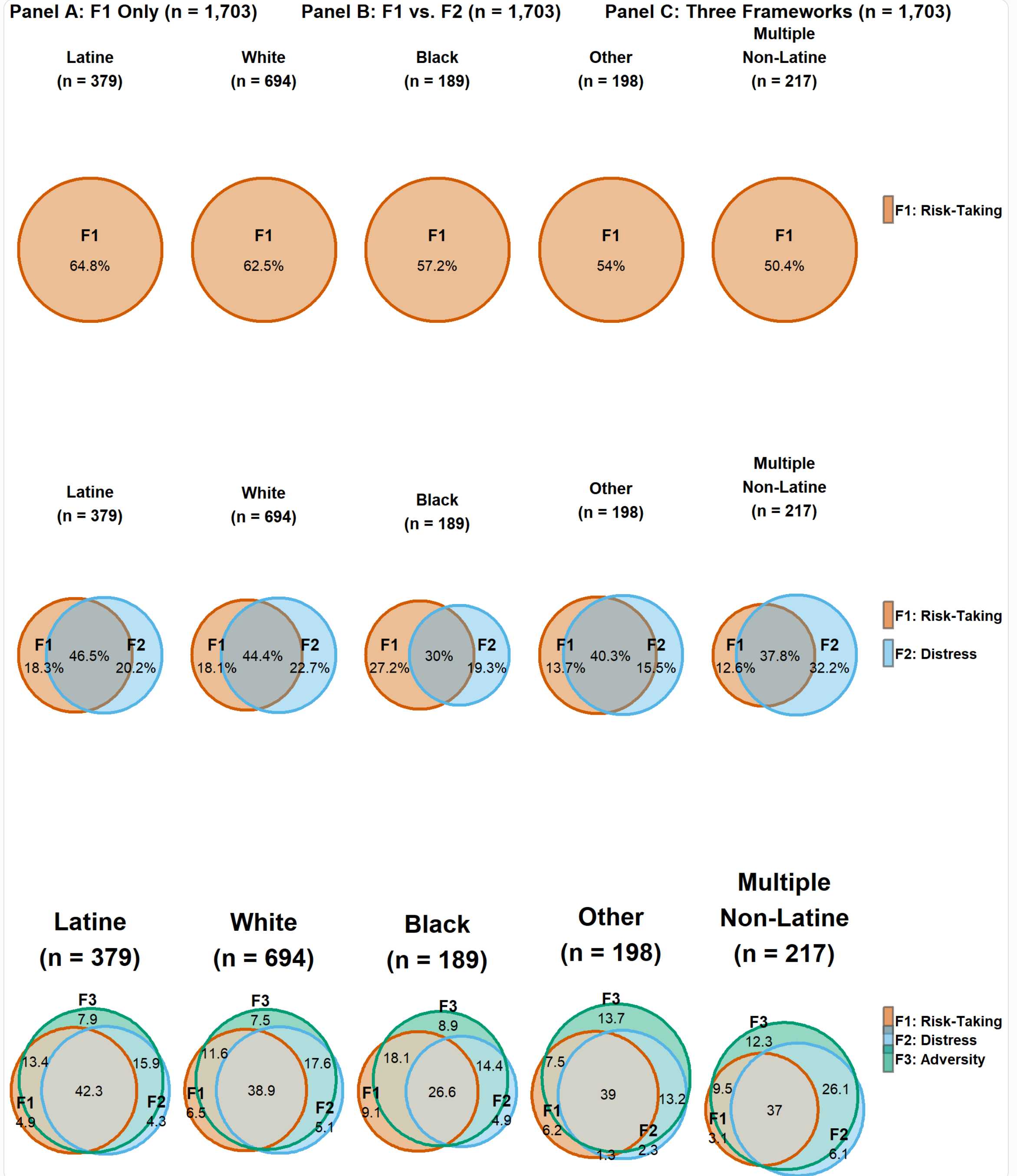
05 HOW DIRECT IS EACH LINK?

Share of each adversity's opioid link carried by general polysubstance use (KHB decomposition). **Lower = more direct, more opioid-specific**.



Bullying's link is almost entirely **direct** (only 10.3% via other drugs); sexual victimization's runs largely through general substance use (25.2%) — a hallmark of broad, not opioid-specific, risk.

06 THE FRAMEWORKS OVERLAP, NOT COMPETE



Reading A—C, the three frameworks collapse into one shared core: for the typical opioid-positive adolescent, risk-taking, distress, and adversity co-occur — the three-way overlap is the largest region in most groups (e.g., 42.3% of Latine). And it is **structural, not incidental**. Among Latine adolescents, nearly all risk-taking + distress also carries contextual adversity (only **4.2%** without it) — acculturative stress, discrimination, economic disadvantage. White adolescents carry somewhat less adversity (greater prescription access); Black adolescents the least (26.6% overlap), consistent with documented racial inequities in opioid prescribing.

07 TAKE-HOME & PREVENTION

Different drivers need different prevention — and because they overlap, the strongest programs layer both.

OPIOID-SPECIFIC
for bullying & suicidality

- Antibullying programs (Sources of Strength)
- Suicide-risk screening (ASQ)
- Distress-tolerance support

GENERAL
for broad adversity & substance use

- Substance screening (CRAFTT, SBIRT)
- Community coalitions (CTC, PROSPER)
- Safe household opioid storage

In clinic, screen both ways: see bullying → think opioids; see opioid use → screen for bullying and unmet mental-health needs.

